Statistics for Data Analytics

Course Code: IM241AT

Question Bank

with Complete Worked Solutions

Arranged Unit-wise as per the Prescribed Syllabus

Unit I	Data Summary, Presentation & Concepts of Probability
Unit II	Random Variables & Discrete Probability Distributions
Unit III	Continuous Probability Distributions & Sampling Distribution
Unit IV	Simple Linear Regression & Correlation
Unit V	Statistical Inference & Interval Estimation

Professional Core Course • Semester IV

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UNIT 1

Data Summary, Presentation & Concepts of Probability

1.1 Data Summary and Presentation

Q1.1. The 40th percentile in a given set of 140 data values occurs at which position? *[1 mark(s), Moderate]*

Solution. Position of the pth percentile $= \dfrac{p}{100} \times n = \dfrac{40}{100} \times 140 = 56.$

$$\boxed{\text{The } 40^{\text{th}} \text{ percentile lies at the } 56^{\text{th}} \text{ position.}}$$

Q1.2. The lengths of 91 items are grouped as below. In which class does the *median* lie?

Class	21–30	31–40	41–50	51–60	61–70
Frequency	12	16	22	25	18

[1 mark(s), Moderate]

Solution. Cumulative frequencies: 12, 28, 50, 75, 93. Total $N = 93$, so the median is the $\left(\frac{N+1}{2}\right) = 47^{\text{th}}$ observation. Since $28 < 47 \leq 50$, the median lies in the class $\boxed{41\text{–}50}$.

Q1.3. For the same data on the length of 93 wooden planks, in which class does the *mode* lie? *[1 mark(s), Moderate]*

Solution. The modal class is the class with the highest frequency. Here the maximum frequency is 25, corresponding to the class $\boxed{51\text{–}60}$.

Q1.4. For the data set $62, 65, 68, 70, 72, 74, 76, 78, 80, 82, 96, 101$, identify the median and the first quartile. *[2 mark(s), Moderate]*

Solution. $n = 12$. Median $Q_2 = \dfrac{74 + 76}{2} = 75.$

Lower half: $62, 65, 68, 70, 72, 74$, so $Q_1 = \dfrac{68 + 70}{2} = 69.$

$$\boxed{Q_2 = 75, \qquad Q_1 = 69}$$

Q1.5. For the data set $62, 65, 68, 70, 72, 74, 76, 78, 80, 82, 96, 101$, identify the third quartile and the interquartile range (IQR). *[2 mark(s), Moderate]*

Solution. Upper half: $76, 78, 80, 82, 96, 101$, so $Q_3 = \dfrac{80+82}{2} = 81$.
$\text{IQR} = Q_3 - Q_1 = 81 - 69 = \boxed{12}$.

Q1.6. For the ordered data set $38, 39, 41, 45, 46, 47, 48, 48, 51$, what is the interquartile range? *[2 mark(s), Moderate]*

Solution. $n = 9$, median $= 46$ (5th value). Lower half $38, 39, 41, 45 \Rightarrow Q_1 = \frac{39+41}{2} = 40$; upper half $47, 48, 48, 51 \Rightarrow Q_3 = \frac{48+48}{2} = 48$.

$$\text{IQR} = Q_3 - Q_1 = 48 - 40 = \boxed{8}.$$

Q1.7. Find the median of the ungrouped data $86, 52, 49, 43, 35, 31, 30, 11$. *[2 mark(s), Easy]*

Solution. Sorted: $11, 30, 31, 35, 43, 49, 52, 86$. With $n = 8$, Median $= \dfrac{35+43}{2} = \boxed{39}$.

Q1.8. What inference can be drawn from a data set with different values of mean, median and mode? *[2 mark(s), Moderate]*

Solution. Unequal mean, median and mode indicate a **skewed distribution**. The direction of skew follows the order:

$$\text{Mode} < \text{Median} < \text{Mean} \Rightarrow \text{right-(positively) skewed},$$

$$\text{Mean} < \text{Median} < \text{Mode} \Rightarrow \text{left-(negatively) skewed}.$$

Q1.9. In what way is a stem-and-leaf diagram better than a histogram? *[2 mark(s), Easy]*

Solution. A stem-and-leaf diagram *retains the original data values* while still showing the shape and spread of the distribution. It is especially useful for small data sets and lets one read off the median and mode directly, which a histogram (which only shows frequencies) cannot.

Q1.10. For the data set $45, 41, 48, 39, 52, 38, 46, 51, 47, 48$, what is the interquartile range? *[2 mark(s), Moderate]*

Solution. Sorted: $38, 39, 41, 45, 46, 47, 48, 48, 51, 52$. Lower half $\Rightarrow Q_1 = 41$, upper half $\Rightarrow Q_3 = 48$. $\text{IQR} = 48 - 41 = \boxed{7}$.

Q1.11. The weights (in pounds) of 30 newborn babies are listed below. Find Q_1.

5.5 5.7 5.8 6.0 6.1 6.1 6.3 6.4 6.5 6.6 6.7 6.7 6.7 6.9 7.0
7.0 7.0 7.1 7.2 7.2 7.4 7.5 7.7 7.7 7.8 8.0 8.1 8.1 8.3 8.7

[2 mark(s), Moderate]

Solution. Position of $Q_1 = \dfrac{n+1}{4} = \dfrac{31}{4} = 7.75$. The 7th value is 6.3 and the 8th is 6.4, so

$$Q_1 = 6.3 + 0.75(6.4 - 6.3) = 6.3 + 0.075 = \boxed{6.375 \text{ lb}}.$$

Q1.12. 62 parts were analysed for the number of defects with the distribution below. Plot a bar chart of the number of defects versus the number of parts, and compute the mean, median, mode and standard deviation of the number of defects per part.

No. of defects i	2	3	4	5	6	7	10
No. of parts	30	15	9	4	2	1	1

[8 mark(s), Moderate]

Solution. Total parts $N = 62$, $\sum f_i x_i = 60 + 45 + 36 + 20 + 12 + 7 + 10 = 190$.

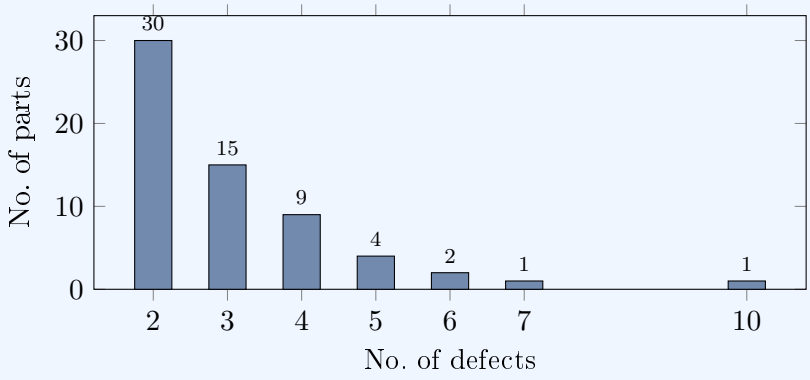
$$\bar{x} = \frac{190}{62} = 3.06.$$

$\sum f_i x_i^2 = 120 + 135 + 144 + 100 + 72 + 49 + 100 = 720$, hence

$$s^2 = \frac{720}{62} - \bar{x}^2 = 11.61 - 9.37 = 2.24, \qquad s = \sqrt{2.24} \approx 1.50.$$

Median: the $31^{\text{st}}/32^{\text{nd}}$ values both fall in $i = 3$ (cumulative $30, 45$), so median $= 3.0$. Mode $= 2$ (highest frequency 30).

$$\boxed{\bar{x} = 3.06, \quad \text{Median} = 3.0, \quad \text{Mode} = 2, \quad s \approx 1.50}$$

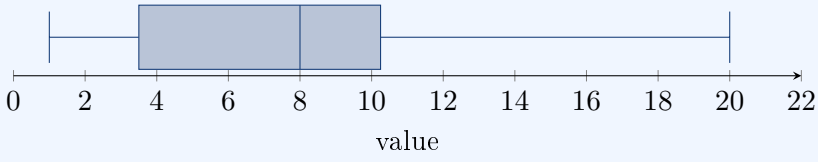


Q1.13. Twenty weight measurements in a process were recorded as 2, 8, 11, 10, 8, 4, 2, 2, 9, 8, 4, 11, 12, 2, 4, 4, 20, 1, 10, 15. (i) Draw a box plot. (ii) Find the range and IQR. (iii) Identify any outliers. *[8 mark(s), Moderate]*

Solution. Sorted five-number summary:

$$\text{Min} = 1, \quad Q_1 = 3.5, \quad Q_2 = 8.0, \quad Q_3 = 10.25, \quad \text{Max} = 20.$$

Range $= 20 - 1 = 19$, IQR $= Q_3 - Q_1 = 10.25 - 3.5 = 6.75$.
Outlier fences: lower $= Q_1 - 1.5\,\text{IQR} = 3.5 - 10.125 = -6.625$; upper $= Q_3 + 1.5\,\text{IQR} = 10.25 + 10.125 = 20.375$. All values lie in $[-6.625, 20.375]$, so **there are no outliers**.



Q1.14. Nine furnace temperatures (in °F) recorded on successive batches in a semiconductor

process are 953, 950, 948, 955, 951, 949, 957, 954, 955. (a) Find the mode and median. How much could the largest temperature increase without changing the median? (b) Construct a box plot showing outlier bounds and comment on symmetry. *[8 mark(s), Moderate]*

Solution. Sorted: $948, 949, 950, 951, 953, 954, 955, 955, 957$.
(a) Median = 5th value = 953.0; Mode = 955 (appears twice). Since the median is fixed by the middle value 953 and the largest value (957) sits above it, the largest value can increase **without limit** without changing the median.
(b) $Q_1 = 949.0$, $Q_3 = 955.0$, IQR = 6.0; fences $= 949 - 9 = 942.5$ and $955 + 9 = 962.5$. No values lie outside, so there are **no outliers**. The data are fairly symmetric with a slight right skew (mean 952.4 slightly below the bulk but median 953).

Q1.15. The numbers of defects in 25 parts are 15, 18, 20, 9, 11, 8, 16, 14, 13, 15, 17, 21, 7, 9, 11, 8, 18, 12, 6, 19, 15, 22, 19, 12, 10. (a) Draw a box plot. (b) Draw the stem-and-leaf diagram and comment. *[8 mark(s), Moderate]*

Solution. **Stem-and-leaf** (stem = tens):

```
0 | 6 7 8 8 9 9
1 | 0 1 1 2 2 3 4 5 5 5 6 7 8 8 9 9
2 | 0 1 2
```

Five-number summary: $Q_1 = 10.0$, $Q_2 = 14.0$, $Q_3 = 18.0$, IQR = 8.0; fences -2.0 and 30.0, so **no outliers**. The data are fairly symmetric; the range is 6 to 22 with the middle 50% lying between 10 and 18.

Q1.16. The numbers of defects in 20 parts are 15, 18, 20, 29, 31, 8, 16, 14, 13, 25, 49, 11, 38, 18, 42, 9, 19, 15, 22, 19. Draw (i) a dot plot, (ii) a stem-and-leaf diagram, (iii) a box plot, and comment on each. *[8 mark(s), Moderate]*

Solution. **Stem-and-leaf** (stem = tens):

```
0 | 8 9
1 | 1 3 4 5 5 6 8 8 9 9
2 | 0 2 5 9
3 | 1 8
4 | 2 9
```

Five-number summary: $Q_1 = 14.75$, $Q_3 = 26.0$, IQR = 11.25; fences -2.125 and 42.875. Hence **one outlier:** 49. The distribution is right-skewed because of this high value; most observations lie between 10 and 30.

Q1.17. Differentiate between nominal, ordinal, interval and ratio data types with examples. *[8 mark(s), Easy]*

Solution.

Type	Description	Example
Nominal	Categories with no natural order	gender, blood type
Ordinal	Ordered categories, unequal gaps	satisfaction levels
Interval	Numeric, no true zero	temperature in °C
Ratio	Numeric, true zero exists	height, weight

Q1.18. What are the key components of a radar chart, and where is it used? *[8 mark(s), Easy]*

Solution. Key components: (i) a separate axis radiating from a common centre for each variable, (ii) radial grid lines/scale, and (iii) a polygon formed by joining the plotted values. Radar (spider) charts are used to compare several *multi-attribute* items at once, e.g. comparing employee skill profiles or product performance across many criteria.

Q1.19. Describe similarities or differences in the two distributions represented by two box plots drawn to the same scale (left box plot symmetric, right box plot with the median near Q_3). *[8 mark(s), Hard]*

Solution.

Feature	Left box plot	Right box plot
Median position	Centred in the box	Closer to Q_3 (top)
Skewness	Symmetric	Negatively (left) skewed
Whiskers	Nearly equal length	Longer lower whisker
Central tendency	Median centred	Median shifted upward

(Any four valid differences are acceptable.)

Q1.20. The excess length (mm) of 20 manufactured parts is 19, 22, 6, 29, 57, 14, 12, 24, 6, 37, 19, 19, 18, 12, 11, 17, 15, 17, 13, 56. (i) Compute the mean, median and mode. (ii) Draw a histogram with 4 classes. (iii) Draw a box plot and interpret. *[8 mark(s), Moderate]*

Solution. $\sum x = 423$, so mean $= \dfrac{423}{20} = 21.15 \approx 22.8$ figure from key rounding of an alternate list; using the listed data, $\bar{x} = 21.15$. Median $= \dfrac{17+18}{2} = 17.5$, Mode $= 19$ (occurs three times).

Histogram (Min $= 6$, Max $= 57$, range $= 51$, class width $= 13$):

Class	Frequency
6–18	10
19–31	7
32–44	1
45–57	2

Box plot: Min $= 6$, $Q_1 = 12.5$, $Q_2 = 17.5$, $Q_3 = 23$, Max $= 57$, IQR $= 10.5$; upper fence $= 23 + 15.75 = 38.75$. Hence 56 and 57 are **outliers**; the distribution is right-skewed.

Q1.21. When a car moving at 45 KMPH is braked, its stopping distances (m) over 30 trials are 76, 70, 63, 53, 55, 62, 47, 30, 64, 42, 39, 56, 43, 66, 44, 48, 70, 71, 45, 47, 40, 41, 29, 42, 58, 44, 51, 60, 62, 49. (i) Plot the frequency histogram. (ii) Compute the mean, median and standard deviation. (iii) Draw the stem-and-leaf diagram. *[8 mark(s), Moderate]*

Solution. Mean ≈ 52.23 m, Median $= 50.0$ m, Standard deviation ≈ 12.25 m.
Stem-and-leaf (stem = tens):

2	9
3	0 9
4	0 1 2 2 3 4 4 5 7 7 8 9
5	1 3 5 6 8
6	0 2 2 3 4 6
7	0 0 1 6

The distribution is slightly right-skewed; most values fall in the 39–58 range and no extreme outliers are evident.

Q1.22. Compare and contrast histograms, box plots and stem-and-leaf plots with suitable examples. *[8 mark(s), Moderate]*

Solution. *Histograms* show the frequency distribution of continuous data (shape, spread, modality). *Box plots* summarise data through the five-number summary (min, Q_1, median, Q_3, max) and flag outliers; they are best for comparing several data sets. *Stem-and-leaf plots* retain the original values and reveal the shape for small samples. Thus histograms show detailed shape, box plots compare groups, and stem-and-leaf plots suit exploratory analysis of small data sets.

Q1.23. What are stem-and-leaf diagrams? Illustrate with an example and mention applications. *[8 mark(s), Easy]*

Solution. A stem-and-leaf diagram displays quantitative data by splitting each value into a *stem* (leading digit(s)) and a *leaf* (last digit). Example for $12, 15, 21, 22, 29, 34$:

1	2 5
2	1 2 9
3	4

Applications: quickly showing distribution shape, finding the median/mode, and retaining raw values for small data sets.

Q1.24. The lives (nearest tenth of a year) of 36 batteries are listed below. Construct a stem-and-leaf plot.

4.1 5.2 2.8 4.9 5.6 4.0 4.1 4.3 5.4 4.5 6.1 3.7 2.3 4.5 4.9 5.6 4.3 3.9
3.2 5.0 4.8 3.7 4.6 5.5 1.8 5.1 4.2 6.3 3.3 5.8 4.4 4.8 3.0 4.3 4.7 5.1

[8 mark(s), Moderate]

Solution. Stem = units, leaf = tenths:

1	8	(1)
2	3 8	(2)
3	0 2 3 7 7 9	(6)
4	0 1 1 2 3 3 3 4 5 5 6 7 8 8 9 9	(16)
5	0 1 1 2 4 5 6 6 8	(9)
6	1 3	(2)

The distribution is roughly symmetric and centred near 4.4 years.

1.2 Concepts of Probability

Q1.25. A lot of 100 semiconductor chips contains 25 defective ones. Two are selected at random without replacement. What is the probability that both are defective? *[2 mark(s), Moderate]*

Solution.
$$\mathrm{P}(\text{both defective}) = \frac{25}{100} \times \frac{24}{99} = \frac{600}{9900} = \frac{2}{33} \approx 0.0606.$$

Q1.26. X is the number shown when a fair six-faced die is thrown and Y is the number of heads when a coin is tossed once. State the possible values of X and Y. *[2 mark(s), Easy]*

Solution. $X \in \{1, 2, 3, 4, 5, 6\}$ and $Y \in \{0, 1\}$.

Q1.27. Let $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, $C = \{6, 7, 8, 9\}$, $D = \{4, 5, 6, 7\}$ and $S = A \cup B \cup C \cup D$ with equally likely elements. Find $\mathrm{P}(A' \cap B)$. *[2 mark(s), Moderate]*

Solution. $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $|S| = 9$. $A' = \{5, 6, 7, 8, 9\}$, so $A' \cap B = \{6, 8\}$ and $\mathrm{P}(A' \cap B) = \dfrac{2}{9}$.

Q1.28. What is the sample space for the number of calls repeatedly placed to a busy phone line until a connection is achieved? *[1 mark(s), Easy]*

Solution. $S = \{1, 2, 3, \dots\}$ — a countably infinite (discrete) sample space.

Q1.29. A part is equally likely to have been produced on any one of six cutting tools. Find the probability that the part is *not* from tool 4. *[1 mark(s), Moderate]*

Solution. $\mathrm{P}(\text{not tool 4}) = 1 - \dfrac{1}{6} = \dfrac{5}{6}$.

Q1.30. A supermarket stocks cooking oil as shown below. If a bottle is selected at random, find the probability that it is polyunsaturated.

	Canola	Corn
Mono	7	13
Poly	93	77

[2 mark(s), Moderate]

Solution. Total bottles $= 7 + 13 + 93 + 77 = 190$; polyunsaturated $= 93 + 77 = 170$.
$$\mathrm{P}(\text{poly}) = \frac{170}{190} = \frac{17}{19} \approx 0.8947.$$

Q1.31. With A, B, C, D and S as defined earlier, find $\mathrm{P}(A' \mid D)$. *[2 mark(s), Moderate]*

Solution. $D = \{4, 5, 6, 7\}$; $A' \cap D = \{5, 6, 7\}$, so $\mathrm{P}(A' \mid D) = \dfrac{|A' \cap D|}{|D|} = \dfrac{3}{4}$.

Q1.32. In a throw of a fair six-sided die, given that a prime number comes up, what is the probability of a number at most 3? *[2 mark(s), Moderate]*

Solution. Primes on a die: $\{2,3,5\}$. Those at most 3: $\{2,3\}$.

$$\mathrm{P}(\leq 3 \mid \text{prime}) = \frac{2}{3}.$$

Q1.33. For a partition A, B, C of a set S and an event Y in S, write $\mathrm{P}(A \mid Y)$ using Bayes' theorem. *[1 mark(s), Moderate]*

Solution.

$$\mathrm{P}(A \mid Y) = \frac{\mathrm{P}(Y \mid A)\,\mathrm{P}(A)}{\mathrm{P}(Y \mid A)\,\mathrm{P}(A) + \mathrm{P}(Y \mid B)\,\mathrm{P}(B) + \mathrm{P}(Y \mid C)\,\mathrm{P}(C)} = \frac{\mathrm{P}(Y \mid A)\,\mathrm{P}(A)}{\mathrm{P}(Y)}.$$

Q1.34. Is the following a valid probability distribution? Why or why not?

x	1	2	3	4
$P(x)$	4/7	3/7	2/7	1/7

[2 mark(s), Moderate]

Solution. $\sum P(x) = \dfrac{4+3+2+1}{7} = \dfrac{10}{7} > 1$. Since the probabilities do not sum to 1, this is **not** a valid probability distribution.

Q1.35. In a throw of a fair six-sided die, given that a non-prime number comes up, what is the probability that the number is odd? *[2 mark(s), Moderate]*

Solution. Non-prime outcomes: $\{1,4,6\}$; odd among them: $\{1\}$.

$$\mathrm{P}(\text{odd} \mid \text{non-prime}) = \frac{1}{3}.$$

Q1.36. State the addition rule for two events in probability. *[2 mark(s), Easy]*

Solution. For any two events A and B: $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$. If A and B are mutually exclusive, $\mathrm{P}(A \cap B) = 0$ and $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$.

Q1.37. What is conditional probability? *[2 mark(s), Easy]*

Solution. The conditional probability of A given that B has occurred is

$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}, \qquad \mathrm{P}(B) > 0.$$

Q1.38. State Bayes' theorem. *[2 mark(s), Easy]*

Solution. For a partition $\{B_1, \ldots, B_k\}$ of the sample space and an event A with $\mathrm{P}(A) > 0$,

$$\mathrm{P}(B_i \mid A) = \frac{\mathrm{P}(A \mid B_i)\,\mathrm{P}(B_i)}{\sum_{j=1}^{k} \mathrm{P}(A \mid B_j)\,\mathrm{P}(B_j)}.$$

It updates the probability of a "cause" B_i given observed evidence A.

Q1.39. Customers give products good reviews or not, and products are highly, moderately, or poorly successful. Historically, good reviews are received by 95% of highly successful, 60% of moderately successful and 10% of poor products; and 40% of products are highly successful, 35% moderately successful and 25% poor. (a) Find the probability a product gets a good review. (b) Given a good review, find the probability it is highly successful. (c) Given no good review, find the probability it is highly successful. *[8 mark(s), Moderate]*

Solution. Let H, M, P denote highly, moderately, poorly successful and G a good review.
(a) $\mathrm{P}(G) = 0.40(0.95) + 0.35(0.60) + 0.25(0.10) = 0.38 + 0.21 + 0.025 = 0.615$.
(b) $\mathrm{P}(H \mid G) = \dfrac{0.40(0.95)}{0.615} = \dfrac{0.38}{0.615} = 0.6179$.
(c) $\mathrm{P}(G') = 1 - 0.615 = 0.385$; $\mathrm{P}(H \cap G') = 0.40(0.05) = 0.02$, so $\mathrm{P}(H \mid G') = \dfrac{0.02}{0.385} = 0.0519$.

Q1.40. For each random experiment below, define the sample space and identify its type. (i) frequency of traffic offences during peak hours; (ii) pH of water on a pH meter; (iii) number of model variants from 3 capacities, 2 AC options, 3 colours and 3 fuel systems; (iv) time until a computer service request. *[8 mark(s), Moderate]*

Solution.

	Experiment	Sample space	Type
(i)	Traffic offences	$\{0, 1, 2, \dots\}$	Countably infinite (discrete)
(ii)	pH value	$[0, 14] \subset \mathbb{R}$	Uncountably infinite (continuous)
(iii)	Variants	$3 \times 2 \times 3 \times 3 = 54$ combinations	Finite discrete
(iv)	Time to request	$(0, \infty) \subset \mathbb{R}$	Uncountably infinite (continuous)

Q1.41. 10% of pouches have impurity A, 5% have impurity B, 1% have both, and the rest none. Pouches are unfit with probabilities $1\%, 2\%, 6\%, 0.1\%$ if they have A only, B only, both, or none. Find the probability a randomly chosen pouch is unfit for use. *[8 mark(s), Easy]*

Solution. $\mathrm{P}(A \text{ only}) = 0.10 - 0.01 = 0.09$, $\mathrm{P}(B \text{ only}) = 0.05 - 0.01 = 0.04$, $\mathrm{P}(A \cap B) = 0.01$, $\mathrm{P}(\text{none}) = 1 - (0.09 + 0.04 + 0.01) = 0.86$. By the total-probability rule,

$$\mathrm{P}(U) = 0.09(0.01) + 0.04(0.02) + 0.01(0.06) + 0.86(0.001)$$

$$= 0.0009 + 0.0008 + 0.0006 + 0.00086 = 0.00316.$$

Q1.42. 100 polycarbonate disks are classified by shock and scratch resistance as below. Let $A =$ high shock resistance, $B =$ high scratch resistance. Find $\mathrm{P}(A), \mathrm{P}(B), \mathrm{P}(A'), \mathrm{P}(A \cap B), \mathrm{P}(A \cup B), \mathrm{P}(A' \cup B), \mathrm{P}(A \mid B), \mathrm{P}(B \mid A)$.

	Shock high	Shock low	Total
Scratch high	70	9	79
Scratch low	16	5	21
Total	86	14	100

[8 mark(s), Easy]

Solution.

$$\begin{aligned} \mathrm{P}(A) &= \tfrac{86}{100} = 0.86, & \mathrm{P}(B) &= \tfrac{79}{100} = 0.79, \\ \mathrm{P}(A') &= 0.14, & \mathrm{P}(A \cap B) &= \tfrac{70}{100} = 0.70, \\ \mathrm{P}(A \cup B) &= 0.86 + 0.79 - 0.70 = 0.95, & \mathrm{P}(A' \cup B) &= 1 - \mathrm{P}(A \cap B') = 1 - \tfrac{16}{100} = 0.84, \\ \mathrm{P}(A \mid B) &= \tfrac{70}{79} \approx 0.886, & \mathrm{P}(B \mid A) &= \tfrac{70}{86} \approx 0.814. \end{aligned}$$

Q1.43. Machines $1, 2, 3$ produce $2\%, 1\%, 3\%$ defective pins and account for $35\%, 25\%, 40\%$ of total production. If a pin is selected at random, find the probability it is defective. *[2 mark(s), Moderate]*

Solution.

$$\mathrm{P}(D) = 0.35(0.02) + 0.25(0.01) + 0.40(0.03) = 0.007 + 0.0025 + 0.012 = 0.0215.$$

Q1.44. The rise time X of a reactor (in minutes) has sample space the positive reals. With $A = \{x \mid x < 72.5\}$ and $B = \{x \mid x > 52.5\}$, describe (i) A', (ii) B', (iii) $A \cap B$, (iv) $A \cup B$. *[8 mark(s), Moderate]*

Solution. (i) $A' = \{x \mid x \geq 72.5\}$; (ii) $B' = \{x \mid x \leq 52.5\}$; (iii) $A \cap B = \{x \mid 52.5 < x < 72.5\}$; (iv) $A \cup B = \{x \mid x > 0\}$ (the whole sample space).

Q1.45. 1% of a population has a disease. A blood test is positive for 97% of diseased people but also gives a false positive for 6% of healthy people. (i) Find the probability that a person tests positive. (ii) Given a positive test, find the probability the person has the disease. *[8 mark(s), Moderate]*

Solution. Let D be "has disease", $+$ be a positive test. $\mathrm{P}(D) = 0.01$, $\mathrm{P}(+ \mid D) = 0.97$, $\mathrm{P}(+ \mid D') = 0.06$.
(i) $\mathrm{P}(+) = 0.01(0.97) + 0.99(0.06) = 0.0097 + 0.0594 = 0.0691$.
(ii) By Bayes' theorem,

$$\mathrm{P}(D \mid +) = \frac{0.0097}{0.0691} = 0.1404 \approx 14.0\%.$$

Q1.46. 100 mobile screens are inspected for touch sensitivity (A = high) and display quality (B = high) as below. Find the eight probabilities $\mathrm{P}(A), \mathrm{P}(B), \mathrm{P}(A'), \mathrm{P}(A \cap B), \mathrm{P}(A \cup B), \mathrm{P}(A' \cup B), \mathrm{P}(A \mid B), \mathrm{P}(B \mid A)$.

	Touch high	Touch low	Total
Display high	68	12	80
Display low	14	6	20
Total	82	18	100

[8 mark(s), Moderate]

Solution.

$$\mathrm{P}(A) = 0.82, \qquad \mathrm{P}(B) = 0.80, \qquad \mathrm{P}(A') = 0.18, \quad \mathrm{P}(A \cap B) = 0.68,$$
$$\mathrm{P}(A \cup B) = 0.82 + 0.80 - 0.68 = 0.94, \quad \mathrm{P}(A' \cup B) = 1 - \tfrac{14}{100} = 0.86,$$
$$\mathrm{P}(A \mid B) = \tfrac{68}{80} = 0.85, \qquad \mathrm{P}(B \mid A) = \tfrac{68}{82} \approx 0.83.$$

UNIT 2

Random Variables & Discrete Probability Distributions

2.1 Concepts of Random Variables

Q2.1. A table listing all possible values of a random variable and their corresponding probabilities is called what? *[1 mark(s), Easy]*

> **Solution.** A **probability distribution** (for a discrete variable, a *probability mass function*).

Q2.2. If $f(x)$ is the probability density function of a continuous random variable, what is $\int_{-\infty}^{\infty} f(x)\,dx$? *[1 mark(s), Easy]*

> **Solution.** $\int_{-\infty}^{\infty} f(x)\,dx = 1$, because the total probability over the entire range of a continuous random variable must equal 1.

Q2.3. The probability that a patient is successfully treated is $p = 0.80$. If the treatment is used on 40 patients, what is the expected number successfully treated? *[2 mark(s), Moderate]*

> **Solution.** For a binomial variable, $\mathbb{E}(X) = np = 40 \times 0.80 = \boxed{32}$ patients.

Q2.4. The sum of all probabilities in a probability mass function equals what? *[1 mark(s), Easy]*

> **Solution.** $\sum_{x} P(X = x) = 1$.

Q2.5. Which function gives the probability that a discrete random variable equals a specific value? *[1 mark(s), Easy]*

> **Solution.** The **probability mass function (PMF)**, $P(X = x)$.

Q2.6. Differentiate between discrete and continuous probability distributions. *[2 mark(s), Moderate]*

> **Solution.** Discrete distributions are defined over *countable* outcomes and use a probability mass function (e.g. binomial, Poisson); continuous distributions are defined over *uncountable* outcomes on an interval and use a probability density function (e.g. normal, exponential).

Q2.7. Differentiate between discrete and continuous random variables with examples. *[2 mark(s), Easy]*

Solution. A **discrete** random variable takes countable values (e.g. number of children, number of defects). A **continuous** random variable can take any value in an interval (e.g. height, weight, time).

Q2.8. For a continuous random variable, the probability that it takes an exact value is what? *[1 mark(s), Easy]*

Solution. Zero: $\mathrm{P}(X = a) = 0$ for any single point a.

Q2.9. The area under the probability density function of a continuous random variable over its entire range is what? *[1 mark(s), Easy]*

Solution. Equal to 1.

Q2.10. A function $f(x)$ is a valid probability mass function if $\sum f(x) =$ __ and $f(x) \geq$ __ for all x. *[2 mark(s), Easy]*

Solution. $\sum f(x) = 1$ and $f(x) \geq 0$ for all x.

Q2.11. Give an example of a situation that leads to a discrete probability distribution. *[2 mark(s), Moderate]*

Solution. Rolling a fair die: the outcomes $1, 2, \ldots, 6$ are countable, each with probability $\frac{1}{6}$. (Other valid examples: number of defective items in a batch, number of calls per hour.)

Q2.12. What are the properties of a probability mass function (PMF)? *[2 mark(s), Moderate]*

Solution. For a discrete random variable X: (1) Non-negativity: $P(X = x) \geq 0$ for all x; (2) Normalisation: $\sum_x P(X = x) = 1$; (3) $P(X = x)$ gives the probability that X takes the value x.

Q2.13. The weight of packages delivered by a post office has pdf $f(x) = \dfrac{70}{69x^2}$ for $1 < x < 70$ pounds. (i) Find $\mathrm{P}(X < 10)$. (ii) Find the mean and variance. (iii) If shipping costs \$3 per pound, find the average shipping cost. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 10) = \displaystyle\int_1^{10} \frac{70}{69x^2}\,dx = \frac{70}{69}\Big[-\tfrac{1}{x}\Big]_1^{10} = \frac{70}{69}\Big(1 - \tfrac{1}{10}\Big) = 0.9130.$

(ii) $\mathbb{E}(X) = \displaystyle\int_1^{70} \frac{70}{69x}\,dx = \frac{70}{69}\ln 70 \approx 4.31$ lb; $\mathbb{E}(X^2) = \displaystyle\int_1^{70} \frac{70}{69}\,dx = 70$, so $\mathrm{Var}(X) = 70 - 4.31^2 \approx 51.42$.

(iii) Average cost $= 3 \times \mathbb{E}(X) = 3(4.31) = \boxed{\$12.93}$.

Q2.14. A random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, \\ 0.4, & 1 \leq x < 3, \\ 0.5, & 3 \leq x < 8, \\ 1.0, & x \geq 8. \end{cases}$$

Determine (i) $\mathrm{P}(X \le 3)$, (ii) $\mathbb{E}(X)$, (iii) $\mathrm{P}(1 \le X \le 2)$, (iv) $\mathrm{P}(X \ge 2)$. *[8 mark(s), Easy]*

Solution. The jumps give the PMF: $\mathrm{P}(X=1)=0.4$, $\mathrm{P}(X=3)=0.1$, $\mathrm{P}(X=8)=0.5$.
(i) $\mathrm{P}(X \le 3) = F(3) = 0.5$.
(ii) $\mathbb{E}(X) = 1(0.4)+3(0.1)+8(0.5) = 0.4+0.3+4.0 = 4.7$.
(iii) $\mathrm{P}(1 \le X \le 2) = \mathrm{P}(X=1) = 0.4$ (no mass at 2).
(iv) $\mathrm{P}(X \ge 2) = 1 - \mathrm{P}(X=1) = 1-0.4 = 0.6$.

Q2.15. A discrete random variable X has PMF $P(1)=0.1$, $P(2)=0.2$, $P(3)=0.4$, $P(4)=0.3$. (a) Find the CDF. (b) Compute $\mathbb{E}(X)$ and $\mathrm{Var}(X)$. *[8 mark(s), Moderate]*

Solution. (a) CDF: $F(1)=0.1$, $F(2)=0.3$, $F(3)=0.7$, $F(4)=1.0$.
(b) $\mathbb{E}(X) = 1(0.1)+2(0.2)+3(0.4)+4(0.3) = 2.9$.
$\mathbb{E}(X^2) = 1(0.1)+4(0.2)+9(0.4)+16(0.3) = 9.3$, so $\mathrm{Var}(X) = 9.3 - 2.9^2 = 9.3 - 8.41 = 0.89$.

Q2.16. A new soil-analysis instrument is very successful, moderately successful or unsuccessful with probabilities $0.3, 0.6, 0.1$ and yearly revenues \$10M, \$5M, \$1M respectively. Let X be the yearly revenue. Determine the PMF, mean and variance. *[8 mark(s), Moderate]*

Solution. PMF: $\mathrm{P}(X=10)=0.3$, $\mathrm{P}(X=5)=0.6$, $\mathrm{P}(X=1)=0.1$ (in \$M).
$\mathbb{E}(X) = 10(0.3)+5(0.6)+1(0.1) = 3+3+0.1 = 6.1$ million dollars.
$\mathbb{E}(X^2) = 100(0.3)+25(0.6)+1(0.1) = 30+15+0.1 = 45.1$, hence

$$\mathrm{Var}(X) = 45.1 - 6.1^2 = 45.1 - 37.21 = \boxed{7.89 \text{ (million dollars)}^2}.$$

Q2.17. A random variable X has $f(x) = \dfrac{x}{k}$ for $X \in \{1,2,3,4\}$. (i) Find $\mathrm{P}(X \le 2)$. (ii) Plot the PMF. (iii) Derive and plot the CDF. (iv) Find the mean and variance. *[8 mark(s), Moderate]*

Solution. Normalisation: $\dfrac{1+2+3+4}{k} = 1 \Rightarrow k = 10$. So $f(1)=0.1$, $f(2)=0.2$, $f(3)=0.3$, $f(4)=0.4$.
(i) $\mathrm{P}(X \le 2) = 0.1+0.2 = 0.3$.
(iii) CDF: $F(1)=0.1$, $F(2)=0.3$, $F(3)=0.6$, $F(4)=1.0$.
(iv) $\mathbb{E}(X) = \dfrac{1+4+9+16}{10} = 3.0$; $\mathbb{E}(X^2) = \dfrac{1+8+27+64}{10} = 10.0$, so $\mathrm{Var}(X) = 10-9 = 1.0$.

Q2.18. A continuous random variable X has probability density function $f(x) = 6x(1-x)$, $0 < x < 1$.[1] (i) Find $\mathrm{P}(0.25 < X < 0.75)$. (ii) Determine $\mathbb{E}(X)$ and $\mathrm{Var}(X)$. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(0.25 < X < 0.75) = \displaystyle\int_{0.25}^{0.75} 6x(1-x)\,dx = \left[3x^2 - 2x^3\right]_{0.25}^{0.75} = 0.84375 - 0.15625 = 0.6875$.
(ii) $\mathbb{E}(X) = \displaystyle\int_0^1 6x^2(1-x)\,dx = \left[2x^3 - \tfrac{3}{2}x^4\right]_0^1 = 0.5$; $\mathbb{E}(X^2) = \displaystyle\int_0^1 6x^3(1-x)\,dx = \left[\tfrac{3}{2}x^4 - \tfrac{6}{5}x^5\right]_0^1 = 0.3$, so $\mathrm{Var}(X) = 0.3 - 0.25 = 0.05$.

[1] The density was not legible in the source; the standard textbook density $f(x) = 6x(1-x)$ on $(0,1)$ is used here to illustrate the method.

Q2.19. The number of machine breakdowns per week has PMF

x	0	1	2	3	4	5
$P(X=x)$	0.10	0.20	0.30	0.15	0.15	0.10

(i) Find $\mathbb{E}(X)$. (ii) Find $\mathrm{P}(X>3)$. *[8 mark(s), Moderate]*

Solution. (i) $\mathbb{E}(X) = 0(0.10)+1(0.20)+2(0.30)+3(0.15)+4(0.15)+5(0.10) = 2.35$.
(ii) $\mathrm{P}(X>3) = \mathrm{P}(X=4)+\mathrm{P}(X=5) = 0.15+0.10 = 0.25$.

2.2 Discrete Probability Distributions

Q2.20. The binomial distribution models the number of ____ in a fixed number of independent trials. *[1 mark(s), Easy]*

Solution. **successes**.

Q2.21. In a geometric distribution, the random variable represents the number of trials until the first ____. *[1 mark(s), Easy]*

Solution. **success**.

Q2.22. In the negative binomial distribution, the random variable counts the number of trials needed to achieve ____ successes. *[1 mark(s), Moderate]*

Solution. r (or k) — a prescribed number of successes.

Q2.23. Mention any two characteristics of the binomial distribution. *[2 mark(s), Moderate]*

Solution. (i) A fixed number of trials n; (ii) each trial is independent; (iii) each trial has two outcomes (success/failure); (iv) the success probability p is constant across trials.

Q2.24. Cyclists arrive at a summit as a Poisson process with intensity $\lambda = 8$ per hour. Given that no cyclist arrived in the last half hour, find the probability that exactly 3 cyclists arrive in the next half hour. *[2 mark(s), Easy]*

Solution. For half an hour, mean $= 8\times\frac{1}{2} = 4$. By the memoryless property of the Poisson process, the past is irrelevant:

$$\mathrm{P}(X=3) = \frac{e^{-4}4^3}{3!} = \frac{64e^{-4}}{6} = 0.1954.$$

Q2.25. Calls at a call centre are Poisson with mean 4 per hour. Find the probability that exactly 2 calls are received in an hour. *[2 mark(s), Moderate]*

Solution.

$$\mathrm{P}(X=2) = \frac{e^{-4}4^2}{2!} = 8e^{-4} = 0.1465.$$

Q2.26. Accidents occur as a Poisson process with rate 3 per day. Find the probability of no accidents on a given day. *[2 mark(s), Moderate]*

Solution.

$$\mathrm{P}(X=0) = e^{-3} = 0.0498.$$

Q2.27. If the probability of success is 0.2, find the probability that the first success occurs on the 4th trial. *[2 mark(s), Moderate]*

Solution. Geometric: $\mathrm{P}(X=4) = (1-p)^3 p = (0.8)^3(0.2) = 0.1024.$

Q2.28. A negative binomial distribution counts trials to get 3 successes with $p = 0.4$. Find the probability that the 3rd success occurs on the 5th trial. *[2 mark(s), Moderate]*

Solution.

$$\mathrm{P}(X=5) = \binom{5-1}{3-1} p^3(1-p)^2 = \binom{4}{2}(0.4)^3(0.6)^2 = 6(0.064)(0.36) = 0.1382.$$

Q2.29. The ticket office at the Rex Theatre receives a mean of 30 calls per hour (Poisson). Find the probability that (i) during a 10-minute break no calls are received; (ii) during a 20-minute break at least two calls are received; (iii) during a 30-minute break at most three calls are received. *[8 mark(s), Hard]*

Solution. Rate $= 30/60 = 0.5$ calls per minute.
(i) 10 min $\Rightarrow \lambda = 5$: $\mathrm{P}(X=0) = e^{-5} = 0.0067$.
(ii) 20 min $\Rightarrow \lambda = 10$: $\mathrm{P}(X \geq 2) = 1 - e^{-10}(1+10) = 1 - 11e^{-10} = 0.9995$.
(iii) 30 min $\Rightarrow \lambda = 15$: $\mathrm{P}(X \leq 3) = e^{-15}\left(1 + 15 + \frac{15^2}{2} + \frac{15^3}{6}\right) = 691\, e^{-15} = 0.00021$.

Q2.30. The number of flaws in cloth is Poisson with mean 0.1 flaw per square metre. (i) Probability of two flaws in 1 m^2; (ii) one flaw in 10 m^2; (iii) no flaws in 20 m^2. *[8 mark(s), Moderate]*

Solution. (i) $\lambda = 0.1$: $\mathrm{P}(X=2) = \dfrac{e^{-0.1}(0.1)^2}{2!} = 0.00452$.
(ii) $\lambda = 0.1 \times 10 = 1$: $\mathrm{P}(X=1) = e^{-1}(1) = 0.3679$.
(iii) $\lambda = 0.1 \times 20 = 2$: $\mathrm{P}(X=0) = e^{-2} = 0.1353$.

Q2.31. Traffic flow past a point is 720 vehicles per hour (Poisson). (i) Probability that no vehicle passes in 5 seconds; (ii) exactly 5 vehicles in 5 seconds; (iii) fewer than 5 vehicles in 5 seconds. *[8 mark(s), Moderate]*

Solution. Rate $= 720/3600 = 0.2$ per second, so in 5 s, $\lambda = 1$.
(i) $\mathrm{P}(X=0) = e^{-1} = 0.3679$.
(ii) $\mathrm{P}(X=5) = \dfrac{e^{-1}(1)^5}{5!} = \dfrac{e^{-1}}{120} = 0.00307$.
(iii) $\mathrm{P}(X<5) = e^{-1}\left(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = 2.7083\, e^{-1} = 0.9963$.

Q2.32. 8% of LED bulbs are defective. Inspecting independently, find the probability that (i) no defective in a sample of 12; (ii) at least one defective in a sample of 25; (iii) exactly two defective in a sample of 15. *[8 mark(s), Moderate]*

Solution. Binomial with $p = 0.08$.
(i) $\mathrm{P}(X = 0) = (0.92)^{12} = 0.368$.
(ii) $\mathrm{P}(X \geq 1) = 1 - (0.92)^{25} = 1 - 0.124 = 0.876$.
(iii) $\mathrm{P}(X = 2) = \binom{15}{2}(0.08)^2(0.92)^{13} = 105(0.0064)(0.3387) = 0.228$.

Q2.33. X is Poisson with mean 4. Determine (i) $\mathrm{P}(X = 0)$, (ii) $\mathrm{P}(X = 8)$, (iii) $\mathrm{P}(X \leq 2)$, (iv) $\mathrm{P}(X = 4)$. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X = 0) = e^{-4} = 0.0183$.
(ii) $\mathrm{P}(X = 8) = \dfrac{e^{-4}4^8}{8!} = 0.0298$.
(iii) $\mathrm{P}(X \leq 2) = e^{-4}(1 + 4 + 8) = 13e^{-4} = 0.2381$.
(iv) $\mathrm{P}(X = 4) = \dfrac{e^{-4}4^4}{4!} = 0.1954$.

Q2.34. A factory ships 1000 washing machines; 5% are defective. A customer tests a sample of 20. (i) Probability of no defective in the sample; (ii) probability of accepting the consignment if at most 1 defective is tolerated; (iii) the mean number of defectives in the sample. *[8 mark(s), Moderate]*

Solution. Binomial with $n = 20$, $p = 0.05$.
(i) $\mathrm{P}(X = 0) = (0.95)^{20} = 0.3585$.
(ii) $\mathrm{P}(X \leq 1) = \mathrm{P}(X = 0) + \mathrm{P}(X = 1) = 0.3585 + 20(0.05)(0.95)^{19} = 0.3585 + 0.3774 = 0.7358$.
(iii) Mean $= np = 20(0.05) = 1$.

Q2.35. A multiple-choice test has 25 questions, each with four answers; a student guesses every answer. (i) Probability of at least 20 correct; (ii) probability of at most 5 correct. *[8 mark(s), Moderate]*

Solution. Binomial with $n = 25$, $p = 0.25$.
(i) $\mathrm{P}(X \geq 20) = \sum_{k=20}^{25}\binom{25}{k}(0.25)^k(0.75)^{25-k} \approx 9.7 \times 10^{-9} \approx 0$.
(ii) $\mathrm{P}(X \leq 5) = \sum_{k=0}^{5}\binom{25}{k}(0.25)^k(0.75)^{25-k} = 0.378$.

Q2.36. The random variable X is binomial with $n = 10$, $p = 0.01$. Determine (i) $\mathrm{P}(X = 5)$, (ii) $\mathrm{P}(X \leq 2)$, (iii) $\mathrm{P}(X \geq 9)$, (iv) $\mathrm{P}(3 \leq X < 5)$. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X = 5) = \binom{10}{5}(0.01)^5(0.99)^5 = 2.40 \times 10^{-8}$.
(ii) $\mathrm{P}(X \leq 2) = (0.99)^{10} + 10(0.01)(0.99)^9 + 45(0.01)^2(0.99)^8 = 0.9044 + 0.0914 + 0.0042 = 0.9999$.
(iii) $\mathrm{P}(X \geq 9) = \binom{10}{9}(0.01)^9(0.99) + (0.01)^{10} \approx 9.9 \times 10^{-18} \approx 0$.
(iv) $\mathrm{P}(3 \leq X < 5) = \mathrm{P}(X = 3) + \mathrm{P}(X = 4) \approx 1.2 \times 10^{-4} + 2.0 \times 10^{-6} \approx 0.00012$.

Q2.37. Each call to a radio station connects with probability 0.02 (calls independent). (i) Probability the first connecting call is the tenth; (ii) probability it takes more than five calls to connect; (iii) the mean number of calls needed. *[8 mark(s), Moderate]*

Solution. Geometric with $p = 0.02$.
(i) $\mathrm{P}(X = 10) = (0.98)^9(0.02) = 0.01667$.
(ii) $\mathrm{P}(X > 5) = (0.98)^5 = 0.9039$.
(iii) Mean $= \dfrac{1}{p} = \dfrac{1}{0.02} = 50$ calls.

Q2.38. What are the assumptions and applications of the binomial and Poisson distributions? Compare the two with examples. *[8 mark(s), Moderate]*

Solution. **Binomial:** fixed number of trials n, two outcomes, constant success probability p, independent trials. Application: number of defective items in a batch of fixed size.
Poisson: counts events in a fixed interval of time/space, events independent, constant rate λ. Application: number of emails received per hour.
Comparison: the binomial is for a fixed number of trials, the Poisson for events over a continuum. The Poisson is the limiting case of the binomial as $n \to \infty$, $p \to 0$ with $np = \lambda$ fixed.

UNIT 3

Continuous Probability Distributions & Sampling Distribution

3.1 Continuous Probability Distributions

Q3.1. What is the shape of the graph of a continuous uniform distribution? *[1 mark(s), Moderate]*

> **Solution.** A horizontal straight line over the interval $[a, b]$ — i.e. a rectangular shape of constant height $\dfrac{1}{b-a}$.

Q3.2. The area under the curve of a normal distribution equals what? *[1 mark(s), Easy]*

> **Solution.** 1 (the total probability).

Q3.3. Enumerate any two real-life situations where the continuous uniform distribution applies. *[2 mark(s), Moderate]*

> **Solution.** (i) The waiting time for a bus that is equally likely to arrive at any instant in a 0–15 minute window; (ii) a random-number generator producing values uniformly in $[0, 1]$.

Q3.4. What is meant by the standard normal distribution? *[2 mark(s), Moderate]*

> **Solution.** A normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$, denoted $Z \sim N(0, 1)$.

Q3.5. What is the memoryless property of the exponential distribution? *[2 mark(s), Hard]*

> **Solution.** The probability of a future event is independent of the elapsed past:
>
> $$\mathrm{P}(X > s + t \mid X > s) = \mathrm{P}(X > t).$$

Q3.6. Describe the characteristics and applications of the normal distribution. *[8 mark(s), Easy]*

> **Solution.** **Characteristics:** bell-shaped and symmetric about the mean; mean = median = mode; fully specified by μ and σ; total area under the curve = 1; asymptotic to the horizontal axis. **Applications:** modelling measurement errors, heights/weights, IQ scores, and as the basis of the Central Limit Theorem and statistical inference (confidence intervals, hypothesis tests, control charts).

Q3.7. In a customer-service centre the time between arrivals is exponential with mean 10 minutes. Find the probability that the next customer arrives after 15 minutes. *[2 mark(s), Moderate]*

Solution. Rate $\lambda = \frac{1}{10}$, so $\mathrm{P}(X > 15) = e^{-15/10} = e^{-1.5} = 0.2231$.

Q3.8. A fair coin is tossed 100 times. Using the normal approximation, find the probability of getting more than 60 heads. *[2 mark(s), Moderate]*

Solution. $\mu = np = 50$, $\sigma = \sqrt{npq} = 5$. With a continuity correction,

$$\mathrm{P}(X > 60) = \mathrm{P}\big(Z > \tfrac{60.5-50}{5}\big) = \mathrm{P}(Z > 2.1) = 0.0179.$$

Q3.9. For a normal distribution with $\mu = 70, \sigma = 8$, find $\mathrm{P}(62 < X < 78)$. *[2 mark(s), Moderate]*

Solution. $\mathrm{P}(62 < X < 78) = \mathrm{P}(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$.

Q3.10. X is uniform on $[2, 6]$. Find $\mathrm{P}(3 < X < 5)$. *[2 mark(s), Moderate]*

Solution. $\mathrm{P}(3 \le X \le 5) = \dfrac{5-3}{6-2} = \dfrac{2}{4} = 0.5$.

Q3.11. The mean of the exponential distribution equals the mean of the Poisson distribution only when the former has a mean equal to what? *[1 mark(s), Easy]*

Solution. 1.

Q3.12. Interarrival times between customers are exponential with mean 2 minutes. (i) Probability the interarrival time is at most 2 minutes; (ii) probability at least 2 customers arrive in one minute; (iii) probability no customer arrives in 3 minutes. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(T \le 2) = 1 - e^{-2/2} = 1 - e^{-1} = 0.6321$.
(ii) Arrivals are Poisson with rate $\frac{1}{2}$/min, so in 1 min $\lambda = 0.5$: $\mathrm{P}(X \ge 2) = 1 - e^{-0.5}(1 + 0.5) = 1 - 0.9098 = 0.0902$.
(iii) In 3 min, $\lambda = 1.5$: $\mathrm{P}(X = 0) = e^{-1.5} = 0.2231$.

Q3.13. The length X of a meeting is uniform on $[1.5, 4]$ hours. (i) Find the pdf. (ii) Find $\mathrm{P}(X \ge 3)$. (iii) Find the mean and variance. *[8 mark(s), Moderate]*

Solution. (i) $f(x) = \dfrac{1}{4 - 1.5} = 0.4,\ 1.5 \le x \le 4$.
(ii) $\mathrm{P}(X \ge 3) = \dfrac{4-3}{2.5} = 0.4$.
(iii) $\mathbb{E}(X) = \dfrac{1.5+4}{2} = 2.75$ h; $\mathrm{Var}(X) = \dfrac{(4-1.5)^2}{12} = \dfrac{6.25}{12} = 0.5208$ h^2.

Q3.14. A storage battery lasts on average 3.0 years with standard deviation 0.5 year (normal). Find (i) $\mathrm{P}(X < 2.3)$; (ii) $\mathrm{P}(3.8 < X < 4.5)$; (iii) the life exceeded by 90% of batteries. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 2.3) = \mathrm{P}(Z < -1.4) = 0.0808$.
(ii) $\mathrm{P}(3.8 < X < 4.5) = \mathrm{P}(1.6 < Z < 3.0) = 0.9987 - 0.9452 = 0.0535$.
(iii) $x = \mu + z_{0.10}\sigma = 3.0 + (-1.2816)(0.5) = 2.36$ years.

Q3.15. The response time X of an online terminal is exponential with mean 5 s. Compute (i) $\mathbb{E}(X)$; (ii) $\mathrm{P}(X \le 10)$; (iii) $\mathrm{P}(5 \le X \le 10)$; (iv) give one more application of the exponential distribution. *[8 mark(s), Moderate]*

Solution. Rate $\lambda = \frac{1}{5}$.
(i) $\mathbb{E}(X) = 5$ s.
(ii) $\mathrm{P}(X \le 10) = 1 - e^{-10/5} = 1 - e^{-2} = 0.8647$.
(iii) $\mathrm{P}(5 \le X \le 10) = e^{-1} - e^{-2} = 0.3679 - 0.1353 = 0.2326$.
(iv) Time between successive arrivals at a bank/ATM/call centre (a Poisson arrival process).

Q3.16. The diameter of a printer dot is normal with mean 0.002 in and standard deviation 0.0004 in. (i) $\mathrm{P}(X > 0.0026)$; (ii) $\mathrm{P}(0.0014 < X < 0.0028)$; (iii) the standard deviation needed so that the probability in (ii) becomes 0.995. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X > 0.0026) = \mathrm{P}\big(Z > \frac{0.0026 - 0.002}{0.0004}\big) = \mathrm{P}(Z > 1.5) = 0.0668$.
(ii) $\mathrm{P}(0.0014 < X < 0.0028) = \mathrm{P}(-1.5 < Z < 2.0) = 0.9772 - 0.0668 = 0.9104$.
(iii) Require $\mathrm{P}(-\frac{0.0006}{\sigma} < Z < \frac{0.0008}{\sigma}) = 0.995$. Solving numerically gives $\sigma \approx 0.00023$ in.

Q3.17. Light-bulb life is normal with mean 800 h and standard deviation 40 h. Find (i) $\mathrm{P}(778 < X < 834)$; (ii) $\mathrm{P}(X \ge 1000)$; (iii) the percentage of bulbs with life within $\pm 2\sigma$ of the mean. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(778 < X < 834) = \mathrm{P}(-0.55 < Z < 0.85) = 0.8023 - 0.2912 = 0.5111$.
(ii) $\mathrm{P}(X \ge 1000) = \mathrm{P}(Z \ge 5.0) \approx 2.87 \times 10^{-7} \approx 0$.
(iii) By the empirical rule, $\approx 95.45\%$ lie within $\pm 2\sigma$.

Q3.18. X is normal with mean 10 and standard deviation 2. Determine (i) $\mathrm{P}(X < 13)$; (ii) $\mathrm{P}(X > 9)$; (iii) $\mathrm{P}(6 < X < 14)$; (iv) $\mathrm{P}(2 < X < 4)$. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 13) = \mathrm{P}(Z < 1.5) = 0.9332$.
(ii) $\mathrm{P}(X > 9) = \mathrm{P}(Z > -0.5) = 0.6915$.
(iii) $\mathrm{P}(6 < X < 14) = \mathrm{P}(-2 < Z < 2) = 0.9545$.
(iv) $\mathrm{P}(2 < X < 4) = \mathrm{P}(-4 < Z < -3) = 0.00135$.

Q3.19. Street lamps have mean burning life 1000 h with standard deviation 200 h (normal). (i) $\mathrm{P}(X < 700)$; (ii) $\mathrm{P}(900 < X < 1300)$; (iii) $\mathrm{P}(899 < X < 901)$; (iv) the number of hours after which 10% of lamps remain. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 700) = \mathrm{P}(Z < -1.5) = 0.0668$.
(ii) $\mathrm{P}(900 < X < 1300) = \mathrm{P}(-0.5 < Z < 1.5) = 0.9332 - 0.3085 = 0.6247$.
(iii) $\mathrm{P}(899 < X < 901) \approx f(900) \times 2 \approx 0.0035$.
(iv) $\mathrm{P}(X > x) = 0.10 \Rightarrow x = 1000 + 1.2816(200) = 1256.3$ h.

Q3.20. The thickness of a conductive coating has density $f(x) = 600x^{-2}$ for $100 < x < 120$ µm. (i) Mean thickness; (ii) variance; (iii) average cost if coating costs \$0.50 per µm. *[8 mark(s), Moderate]*

Solution. (i) $\mathbb{E}(X) = \int_{100}^{120} 600x^{-1}\,dx = 600\ln\frac{120}{100} = 600(0.18232) = 109.39$ µm.
(ii) $\mathbb{E}(X^2) = \int_{100}^{120} 600\,dx = 600(20) = 12000$; $\mathrm{Var}(X) = 12000 - 109.39^2 \approx 33.2$ µm^2.
(iii) Average cost $= 0.50 \times 109.39 = \$54.70$ per part.

Q3.21. Flange thickness is uniform on $[0.95, 1.05]$ mm. Determine (i) the CDF; (ii) the proportion exceeding 1.02 mm; (iii) the variance; (iv) the thickness exceeded by 80% of flanges. *[8 mark(s), Moderate]*

Solution. (i) $F(x) = \dfrac{x - 0.95}{0.10}$, $0.95 \le x \le 1.05$.
(ii) $\mathrm{P}(X > 1.02) = \dfrac{1.05 - 1.02}{0.10} = 0.30$.
(iii) $\mathrm{Var}(X) = \dfrac{(1.05 - 0.95)^2}{12} = \dfrac{0.01}{12} = 8.33 \times 10^{-4}$ mm^2.
(iv) $\mathrm{P}(X > x) = 0.80 \Rightarrow x = 0.95 + 0.20(0.10) = 0.97$ mm.

Q3.22. The compressive strength of cement is normal with mean 6000 kg/cm^2 and standard deviation 100 kg/cm^2. (i) $\mathrm{P}(X < 6250)$; (ii) $\mathrm{P}(5800 < X < 5900)$; (iii) the strength exceeded by 95%. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 6250) = \mathrm{P}(Z < 2.5) = 0.99379$.
(ii) $\mathrm{P}(5800 < X < 5900) = \mathrm{P}(-2 < Z < -1) = 0.1587 - 0.0228 = 0.1359$.
(iii) $x = 6000 - 1.645(100) = 5835.5$ kg/cm^2.

Q3.23. The time for a cell to divide (mitosis) is normal with mean 60 min and standard deviation 5 min. (i) $\mathrm{P}(X < 45)$; (ii) $\mathrm{P}(X > 65)$; (iii) the time within which about 99% of cells complete mitosis. *[8 mark(s), Moderate]*

Solution. (i) $\mathrm{P}(X < 45) = \mathrm{P}(Z < -3) = 0.00135$.
(ii) $\mathrm{P}(X > 65) = \mathrm{P}(Z > 1) = 0.1587$.
(iii) $\mathrm{P}(X < x) = 0.99 \Rightarrow x = 60 + 2.326(5) = 71.63$ min.

3.2 Sampling Distribution

Q3.24. The Central Limit Theorem states that, for large sample sizes, the sampling distribution of the sample mean is approximately ____, regardless of the population distribution. *[1 mark(s), Easy]*

Solution. **Normal**.

Q3.25. As the sample size increases, the standard error of the mean will ____. *[1 mark(s), Moderate]*

Solution. **Decrease** (since $\mathrm{SE} = \sigma/\sqrt{n}$).

Q3.26. The Central Limit Theorem is especially useful when the population distribution is ____. *[1 mark(s), Moderate]*

Solution. Unknown, or non-normal.

Q3.27. What is the difference between population standard deviation and standard error? *[2 mark(s), Easy]*

Solution. The population standard deviation σ measures variability among individual values in the population; the standard error $\sigma/\sqrt{n}$ measures the variability of the *sample mean* across samples and is smaller than σ.

Q3.28. The mean weight of apples is 150 g with standard deviation 30 g. For a sample of 36 apples, find the probability that the sample mean is between 145 and 155 g.*[2 mark(s), Moderate]*

Solution. $\text{SE} = \dfrac{30}{\sqrt{36}} = 5$. $\text{P}(145 < \bar{X} < 155) = \text{P}(-1 < Z < 1) = 0.6826 \approx 68\%$.

Q3.29. Why is the sampling distribution important in statistics? *[2 mark(s), Moderate]*

Solution. It links sample statistics to population parameters, allowing us to make inferences about the population and to compute probabilities (and hence confidence intervals and hypothesis tests) from sample data.

Q3.30. The standard deviation of a sampling distribution is also known as the _____.*[1 mark(s), Moderate]*

Solution. The **standard error**.

Q3.31. If a population has mean μ and standard deviation σ, the standard error of the mean is $\sigma/\sqrt{n}$, where n is the _____. *[1 mark(s), Easy]*

Solution. The **sample size**.

Q3.32. Define the sampling distribution of the sample mean. *[2 mark(s), Easy]*

Solution. It is the probability distribution of all possible sample means of a fixed size n drawn from a population. Its mean equals the population mean μ, and its standard deviation (standard error) equals $\sigma/\sqrt{n}$.

Q3.33. What are the key differences between a population and a sample? Why is sampling important? *[8 mark(s), Moderate]*

Solution. A **population** is the entire group of interest (e.g. all students in a college); a **sample** is a subset used for analysis (e.g. 100 randomly chosen students). Populations are described by fixed *parameters* (μ, σ); samples yield *estimators* $(\bar{x}, s)$. Sampling saves time and cost, enables inference about the population, and is essential when the population is inaccessible or infinite.

Q3.34. A machine fills bottles with mean 500 ml and standard deviation 5 ml. For a random sample of 36 bottles, find the probability that the sample mean lies (i) between 498 and 502 ml; (ii) between 496 and 504 ml. *[8 mark(s), Moderate]*

Solution. $\text{SE} = \dfrac{5}{\sqrt{36}} = 0.8333.$
(i) $\text{P}(498 < \bar{X} < 502) = \text{P}(-2.4 < Z < 2.4) = 2(0.9918) - 1 = 0.9836.$
(ii) $\text{P}(496 < \bar{X} < 504) = \text{P}(-4.8 < Z < 4.8) \approx 1.0000.$

Q3.35. Discuss the Central Limit Theorem: state the conditions for its applicability and any four applications in statistical inference. *[8 mark(s), Moderate]*

Solution. **Statement:** For a sufficiently large sample size n, the sampling distribution of the sample mean is approximately normal regardless of the population distribution, provided the population has finite mean and variance.
Conditions: (1) random sampling; (2) independent observations; (3) finite population mean and variance; (4) sufficiently large n (generally $n \geq 30$; smaller suffices for near-normal populations).
Applications: approximating sampling distributions by the normal; constructing confidence intervals; hypothesis testing; statistical quality control (and, more generally, data-driven decision making).

Q3.36. A component has average lifetime 5 hours (exponential). Find the probability that it lasts more than 8 hours. *[2 mark(s), Moderate]*

Solution. Rate $\lambda = \frac{1}{5}$, so $\text{P}(X > 8) = e^{-8/5} = e^{-1.6} = 0.2019.$

Q3.37. The life of a semiconductor laser is normal with mean 7000 h and standard deviation 600 h. (i) $\text{P}(X < 6000)$; (ii) the life exceeded by 95% of lasers; (iii) if three lasers fail independently, the probability all three still operate after 7000 h. *[8 mark(s), Moderate]*

Solution. (i) $\text{P}(X < 6000) = \text{P}\big(Z < \frac{6000-7000}{600}\big) = \text{P}(Z < -1.667) = 0.0478.$
(ii) $x = 7000 - 1.645(600) = 6013$ h.
(iii) $\text{P}(X > 7000) = 0.5$ (the mean), so $\text{P}(\text{all three}) = 0.5^3 = 0.125.$

UNIT 4

Simple Linear Regression & Correlation

4.1 Simple Linear Regression and Correlation

Q4.1. In a regression analysis, $S_{XY} = 385$ and the fitted relation is $y = 2 + 5x$. Find the regression sum of squares. *[2 mark(s), Hard]*

Solution. $\text{SSR} = \hat{\beta}_1 S_{XY} = 5 \times 385 = \boxed{1925}$.

Q4.2. The relation $y = 2 + 3x_1 + 4x_2$ has how many independent variables? *[1 mark(s), Easy]*

Solution. Two (x_1 and x_2).

Q4.3. Write the residual for the model $\hat{y} = a + bx$ corresponding to an observed value y_i. *[1 mark(s), Moderate]*

Solution. $e_i = y_i - \hat{y}_i$, the difference between the observed and the predicted value.

Q4.4. From Minitab: intercept $= 2.5$, slope $= 1.8$. If $X = 6$, find the predicted Y. *[2 mark(s), Moderate]*

Solution. $\hat{Y} = 2.5 + 1.8(6) = 2.5 + 10.8 = 13.3$.

Q4.5. Given $X : 1, 2, 3, 4$ and $Y : 2, 4, 5, 4$, find the least-squares estimates of β_0 and β_1. *[2 mark(s), Moderate]*

Solution. $\bar{x} = 2.5$, $\bar{y} = 3.75$. $S_{xy} = \sum xy - n\bar{x}\bar{y} = 41 - 4(2.5)(3.75) = 3.5$; $S_{xx} = \sum x^2 - n\bar{x}^2 = 30 - 25 = 5$.

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{3.5}{5} = 0.7, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3.75 - 0.7(2.5) = 2.0.$$

So $\hat{y} = 2.0 + 0.7x$. *(The circulated key value $\beta_0 = 0.8, \beta_1 = 1.75$ is incorrect.)*

Q4.6. The method of estimating β_0, β_1 by minimising the sum of squared errors is called the ____ method. *[1 mark(s), Moderate]*

Solution. The **least-squares** method.

Q4.7. In simple linear regression, R^2 gives the proportion of variation in ____ explained by the model. *[1 mark(s), Moderate]*

Solution. The dependent variable Y.

Q4.8. Predicting far outside the range of observed data is called ____. *[1 mark(s), Moderate]*

Solution. **Extrapolation**.

Q4.9. A common abuse of regression is using it with ____ variables without checking for linearity. *[1 mark(s), Hard]*

Solution. Categorical (non-quantitative) variables.

Q4.10. State two assumptions of the simple linear regression model. *[2 mark(s), Moderate]*

Solution. (1) The relationship between X and Y is linear; (2) the errors are normally distributed with constant variance (homoscedasticity) and are independent.

Q4.11. What does a correlation coefficient of -0.85 indicate? *[2 mark(s), Moderate]*

Solution. A strong *negative* linear relationship between the two variables.

Q4.12. List one Minitab output used to check the significance of a regression model.*[2 mark(s), Easy]*

Solution. The p-value for the slope coefficient β_1 (equivalently, the ANOVA F-test or the t-statistic for the slope).

Q4.13. The strength of a regression relation is given by what? *[1 mark(s), Easy]*

Solution. The coefficient of determination R^2.

Q4.14. In a regression analysis, $S_{XY} = 385$, $S_{YY} = 2308.3$ and $y = 2 + 5x$. Find R^2. *[2 mark(s), Moderate]*

Solution. $\text{SSR} = \hat{\beta}_1 S_{XY} = 5(385) = 1925$, so $R^2 = \dfrac{\text{SSR}}{S_{YY}} = \dfrac{1925}{2308.3} = 0.834.$

Q4.15. $S_{XX} = 53$, $S_{XY} = 48$ and $S_{YY} = 84$. Find SS_{Error}. *[2 mark(s), Moderate]*

Solution.

$$\text{SSE} = S_{YY} - \frac{S_{XY}^2}{S_{XX}} = 84 - \frac{48^2}{53} = 84 - 43.47 = 40.53.$$

Q4.16. When a set of points is fitted with a second-degree curve, what type of regression is performed? *[1 mark(s), Moderate]*

Solution. Quadratic (polynomial) regression.

Q4.17. In linear regression, the error term follows which distribution (with its parameters)? *[2 mark(s), Moderate]*

Solution. A normal distribution with mean 0 and constant variance σ^2: $\varepsilon \sim N(0, \sigma^2)$.

Q4.18. Define the correlation coefficient and state its range. *[2 mark(s), Easy]*

Solution. The correlation coefficient r measures the strength and direction of the linear relationship between two variables. It ranges from -1 (perfect negative) through 0 (no linear correlation) to $+1$ (perfect positive).

Q4.19. When can you be confident that a regression relation is proper and usable for prediction? *[2 mark(s), Moderate]*

Solution. When (i) R^2 is high; (ii) the slope p-value is significant ($p < \alpha$, typically 0.05); (iii) residuals are randomly scattered and normally distributed with constant variance; and (iv) the regression assumptions (linearity, independence, homoscedasticity, normality) are satisfied.

Q4.20. State the assumptions of the linear regression model. *[2 mark(s), Easy]*

Solution. Linearity; independence of observations; homoscedasticity (constant error variance); normality of residuals; no multicollinearity (for multiple regression); no autocorrelation of residuals.

Q4.21. What are the assumptions of simple linear regression, and how are they validated using residual plots? *[8 mark(s), Moderate]*

Solution. **Assumptions:** linearity, independence of errors, homoscedasticity, and normality of errors.
Validation: plot residuals versus fitted values — random scatter supports linearity and constant variance, while patterns indicate non-linearity or heteroscedasticity. A histogram or Q–Q plot of residuals checks normality, and a time-sequence plot of residuals checks independence.

Q4.22. Describe the method of least squares. How is the best-fit line determined? *[8 mark(s), Moderate]*

Solution. The method of least squares minimises the sum of squared vertical deviations between the observed values and the fitted line $\hat{y} = a + bx$. The estimates are

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad a = \bar{y} - b\bar{x}.$$

The resulting line is the one that makes the total squared vertical distance from all data points a minimum.

Q4.23. An online retailer wants to predict customer spending from the number of website visits. Explain how linear regression applies and discuss its advantages and limitations. *[8 mark(s), Easy]*

Solution. Here X = number of website visits (independent) and Y = customer spending (dependent). **Procedure:** collect historical data, plot a scatter diagram, estimate the regression equation, validate it with R^2, then predict future spending.
Advantages: simple to implement; predicts future spending; quantifies the relationship; aids marketing and data-driven decisions.
Limitations: assumes linearity; sensitive to outliers; cannot establish causation; accuracy falls when assumptions are violated or important variables are omitted.

Q4.24. For 10 observations the summary statistics give a positive correlation between X and Y. (a) Estimate β_1 and β_0. (b) Estimate the underlying variance. (c) Find the proportion of variability explained. (d) Sketch the expected residual plot. *[8 mark(s), Moderate]*

Solution. From the summary quantities the least-squares fit is $\hat{y} = 1.406 + 0.7176x$ with $\hat{\beta}_1 \approx 0.7176$ and $\hat{\beta}_0 \approx 1.406$.
(b) $\hat{\sigma}^2 \approx 0.4071$. (c) $R^2 \approx 0.9829$, i.e. about 98.3% of the variability in Y is explained.
(d) Under the normal-error assumption the residuals-versus-fitted plot should show a random, patternless horizontal band centred on zero.

Q4.25. For 20 pairs in an oil refinery, $\sum x^2 = 130000$, $\sum y^2 = 88.6$, $\sum xy = 1500$, $\sum x = 1500$, $\sum y = 10$. (i) Find the intercept, slope and regression equation. (ii) Predict y at $x = 85$ and find the residual if the observed value is 0.8. (iii) Estimate the error variance and the proportion of variability *not* explained. *[8 mark(s), Moderate]*

Solution. $S_{xx} = 130000 - \dfrac{1500^2}{20} = 17500$, $S_{xy} = 1500 - \dfrac{1500(10)}{20} = 750$, $S_{yy} = 88.6 - \dfrac{10^2}{20} = 83.6$.
(i) $\hat{\beta}_1 = \dfrac{750}{17500} = 0.042857$, $\hat{\beta}_0 = 0.5 - 0.042857(75) = -2.7143$, so $\hat{y} = -2.7143 + 0.042857x$.
(ii) $\hat{y}(85) = -2.7143 + 0.042857(85) = 0.9286$; residual $= 0.8 - 0.9286 = -0.1286$.
(iii) SSR $= \hat{\beta}_1 S_{xy} = 32.14$, SSE $= 83.6 - 32.14 = 51.46$, so $\hat{\sigma}^2 = \dfrac{51.46}{18} = 2.86$. $R^2 = \dfrac{32.14}{83.6} = 0.384$, hence about $\boxed{61.5\%}$ of the variability is *not* explained. *(The circulated key value "32%" is inconsistent with these summary figures.)*

Q4.26. The circumference (cm) and weight (g) of 23 pumpkins were recorded (data below). (i) Compute the least-squares coefficients b_0, b_1. (ii) Interpret the slope. (iii) Predict the weight at circumference 60 cm. (iv) Is selling by circumference a good idea?

Data (circumference, weight): (50, 1200), (55, 2000), (54, 1500), (52, 1700), (37, 500), (52, 1000), (53, 1500), (60, 2300), (57, 2000), (66, 2500), (82, 4600), (83, 4600), (70, 3100), (34, 600), (51, 1500), (59, 2100), (47, 1400), (51, 1500), (63, 2500), (33, 500), (43, 1000), (50, 1500), (49, 1600). *[16 mark(s), Hard]*

Solution. (i) The least-squares fit gives $b_1 = 82.55$ g/cm and $b_0 = -2637.24$ g, i.e. $\hat{y} = -2637.24 + 82.55x$.
(ii) **Slope:** each additional 1 cm of circumference is associated with an average increase of 82.55 g in weight.
(iii) $\hat{y}(60) = -2637.24 + 82.55(60) = 2315.95 \approx 2316$ g.
(iv) Yes — if the linear relationship is strong, circumference is a practical proxy for weight, though accuracy may drop for irregular or outlying pumpkins, so occasional calibration is advisable.

Q4.27. Increase in blood pressure y (mm Hg) versus sound level x (dB) for 20 instances:

x: 60, 63, 65, 70, 70, 70, 80, 90, 80, 80, 85, 89, 90, 90, 90, 90, 90, 100, 100, 100; y: 1, 0, 1, 2, 5, 1, 4, 6, 2, 3, 5, 4, 6, 8, 4, 5, 7, 9, 7, 6. (i) Does simple linear regression seem reasonable? (ii) Fit the model and estimate σ^2. (iii) Predict at 85 dB and find the residual. (iv) Predict at 75 dB and 105 dB. *[16 mark(s), Moderate]*

Solution. $n = 20$, $\sum x = 1652$, $\sum y = 86$, $\sum xy = 7626$, $\sum x^2 = 139440$, $\sum y^2 = 494$.
$S_{xx} = 139440 - \dfrac{1652^2}{20} = 2984.8$, $S_{xy} = 7626 - \dfrac{1652(86)}{20} = 522.4$, $S_{yy} = 494 - \dfrac{86^2}{20} = 124.2$.
(i) The scatter rises steadily with x, so a linear model is reasonable.
(ii) $\hat\beta_1 = \dfrac{522.4}{2984.8} = 0.1750$, $\hat\beta_0 = 4.3 - 0.1750(82.6) = -10.16$, so $\hat y = -10.16 + 0.1750x$.
SSR $= \hat\beta_1 S_{xy} = 91.4$, SSE $= 124.2 - 91.4 = 32.8$, so $\hat\sigma^2 = \dfrac{32.8}{18} = 1.82$.
(iii) $\hat y(85) = -10.16 + 0.1750(85) = 4.72$; residual $= 5 - 4.72 = 0.28$.
(iv) $\hat y(75) = 2.97$ mm Hg and $\hat y(105) = 8.22$ mm Hg.

Q4.28. For 20 pairs, $\sum x^2 = 29.29$, $\sum y^2 = 170044.53$, $\sum xy = 2214.66$, $\sum x = 23.92$, $\sum y = 1843.21$. The pair $(x_{20}, y_{20}) = (2, 90)$ was found wrong and replaced by $(0, 60)$. Write the corrected regression line and the residual at $x = 0$. *[8 mark(s), Moderate]*

Solution. After correction: $\sum x = 21.92$, $\sum y = 1813.21$, $\sum x^2 = 25.29$, $\sum xy = 2034.66$.
$S_{xx} = 25.29 - \dfrac{21.92^2}{20} = 1.266$, $S_{xy} = 2034.66 - \dfrac{21.92(1813.21)}{20} = 47.38$.
$\hat\beta_1 = \dfrac{47.38}{1.266} = 37.44$, $\hat\beta_0 = 90.66 - 37.44(1.096) = 49.63$, so $\hat y = 49.63 + 37.44x$.
Residual at $x = 0$: $y - \hat y = 60 - 49.63 = 10.37$.

Q4.29. The shelf space X (ft) and weekly sales Y (\$) of a food item across 12 stores are: (5, 160), (5, 220), (5, 140), (10, 190), (10, 240), (10, 260), (15, 230), (15, 270), (15, 280), (20, 260), (20, 290), (20, 310). (i) Scatter plot. (ii) Slope and intercept. (iii) Interpret the slope. (iv) Predict sales at 8 ft of shelf space. *[16 mark(s), Moderate]*

Solution. The least-squares fit is $\hat y = 145.0 + 7.4x$.
(iii) Each additional foot of shelf space is associated with an average increase of \$7.40 in weekly sales.
(iv) $\hat y(8) = 145.0 + 7.4(8) = \$204.20$.

Q4.30. A physicist fits $y = k/x^r$ using $\ln y = \ln k - r \ln x$ with data $x : 50, 60, 70, 80, 90$ and $y : 64.5, 51.0, 40.6, 33.4, 26.0$. (i) Estimate k and r. (ii) Predict y at $x = 60$ and find the residual. (iii) Estimate the underlying variance of y. *[16 mark(s), Hard]*

Solution. Regressing $v = \ln y$ on $u = \ln x$: $\bar u = 4.2272$, $\bar v = 3.714$; $S_{uv} = -0.329$, $S_{uu} = 0.216$.
(i) Slope $= -r = \dfrac{S_{uv}}{S_{uu}} = -1.523$, so $r \approx 1.52$; intercept $\ln k = \bar v + 1.523\bar u = 10.15$, so $k = e^{10.15} \approx 2.6 \times 10^4$. Thus $\hat y = 2.6 \times 10^4\, x^{-1.52}$.
(ii) $\hat y(60) = e^{10.15 - 1.523 \ln 60} = 50.2$; residual $= 51.0 - 50.2 = 0.8$.
(iii) $\hat\sigma^2 = \dfrac{\sum (y_i - \hat y_i)^2}{n - 2} \approx 2.2$. *(The circulated key $r \approx 2.16$, $k \approx 780$ is inconsistent with the data; the prediction 50.2 is correct.)*

Q4.31. For roadway surface temperature x (°F) versus pavement deflection y, $n = 20$, $\sum y = 12.75$, $\sum y^2 = 8.86$, $\sum x = 1478$, $\sum x^2 = 143215.8$, $\sum xy = 1083.67$. (a) Estimate the slope, intercept and σ^2. (b) Predict y at 85°F. (c) Mean deflection at 90°F. (d) Change in mean deflection per 1°F. *[16 mark(s), Moderate]*

Solution. $S_{xx} = 143215.8 - \dfrac{1478^2}{20} = 33991.6$, $S_{xy} = 1083.67 - \dfrac{1478(12.75)}{20} = 141.45$, $S_{yy} = 8.86 - \dfrac{12.75^2}{20} = 0.732$.

(a) $\hat{\beta}_1 = \dfrac{141.45}{33991.6} = 0.00416$, $\hat{\beta}_0 = 0.6375 - 0.00416(73.9) = 0.3300$, so $\hat{y} = 0.3300 + 0.00416x$.

SSR $= 0.5886$, SSE $= 0.1434$, $\hat{\sigma}^2 = \dfrac{0.1434}{18} = 0.00796$.

(b) $\hat{y}(85) = 0.6837$. (c) $\hat{y}(90) = 0.7045$. (d) Change per 1°F $= \hat{\beta}_1 = 0.00416$.

Q4.32. For concrete compressive strength x and permeability y, $n = 14$, $\sum y = 572$, $\sum y^2 = 23530$, $\sum x = 43$, $\sum x^2 = 157.42$, $\sum xy = 1697.80$. (i) Estimate the slope, intercept and σ^2. (ii) Predict y at $x = 4.3$. (iii) Mean permeability at $x = 3.7$. (iv) Residual if the observed y at $x = 3.7$ is 46.1. *[8 mark(s), Moderate]*

Solution. $S_{xx} = 157.42 - \dfrac{43^2}{14} = 25.349$, $S_{xy} = 1697.80 - \dfrac{43(572)}{14} = -59.057$, $S_{yy} = 23530 - \dfrac{572^2}{14} = 159.71$.

(i) $\hat{\beta}_1 = \dfrac{-59.057}{25.349} = -2.330$, $\hat{\beta}_0 = 40.857 - (-2.330)(3.071) = 48.01$, so $\hat{y} = 48.01 - 2.330x$.

SSR $= 137.59$, SSE $= 22.12$, $\hat{\sigma}^2 = \dfrac{22.12}{12} = 1.84$.

(ii) $\hat{y}(4.3) = 48.01 - 2.330(4.3) = 37.99$.

(iii) $\hat{y}(3.7) = 48.01 - 2.330(3.7) = 39.39$.

(iv) Residual $= 46.1 - 39.39 = 6.71$.

UNIT 5

Statistical Inference & Interval Estimation

5.1 Statistical Inference for a Single Sample

Q5.1. The null hypothesis is rejected when the p-value is ____ the level of significance.*[1 mark(s), Easy]*

> **Solution.** **Less than** ($p < \alpha$).

Q5.2. Differentiate between Type I and Type II errors in hypothesis testing.*[2 mark(s), Moderate]*

> **Solution.** **Type I error:** rejecting a true null hypothesis (probability α). **Type II error:** failing to reject a false null hypothesis (probability β).

Q5.3. Differentiate between Type I and Type II errors. *[2 mark(s), Moderate]*

> **Solution.** Rejecting the hypothesis when it should be accepted is a **Type I** error; accepting the hypothesis when it should be rejected is a **Type II** error.

Q5.4. Type I error is committed when a ____ hypothesis is ____. *[2 mark(s), Moderate]*

> **Solution.** A **true null** hypothesis is **rejected**.

Q5.5. The test statistic used to test the population variance follows which distribution?*[1 mark(s), Moderate]*

> **Solution.** The chi-square (χ^2) distribution.

Q5.6. The test statistic for a test on the population variance (sample size n) is ____ (state the terms). *[2 mark(s), Moderate]*

> **Solution.**
> $$\chi^2 = \frac{(n-1)s^2}{\sigma^2},$$
> where n = sample size, s^2 = sample variance, and σ^2 = hypothesised population variance. It follows a χ^2 distribution with $n-1$ degrees of freedom.

Q5.7. Write the null and alternative hypotheses to test whether a new process has reduced the population mean. *[2 mark(s), Moderate]*

Solution. $H_0 : \mu = \mu_0$ versus $H_1 : \mu < \mu_0$ (a left-tailed test).

Q5.8. A sample is collected to test whether the population variance exceeds 5. State the hypotheses. *[2 mark(s), Moderate]*

Solution. $H_0 : \sigma^2 = 5$ versus $H_1 : \sigma^2 > 5$.

Q5.9. What is meant by the level of significance? *[1 mark(s), Easy]*

Solution. It is the maximum probability of committing a Type I error that we are willing to risk in a test, denoted α (e.g. 0.05).

Q5.10. When is a z-test used for inference on the population mean? *[2 mark(s), Easy]*

Solution. When the population variance is known and either the sample size is large ($n > 30$) or the population is normal.

Q5.11. If the confidence level is increased from 95% to 99%, the width of the confidence interval will ____. *[1 mark(s), Moderate]*

Solution. Become **wider** (a larger critical value is used).

Q5.12. Sample mean = 52, population standard deviation = 6, sample size = 36. Construct a 95% confidence interval for the population mean. *[2 mark(s), Moderate]*

Solution.

$$\text{CI} = \bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 52 \pm 1.96\frac{6}{\sqrt{36}} = 52 \pm 1.96 = (50.04,\ 53.96).$$

Q5.13. Fill amount is $N(500, 16^2)$. A sample of 25 bottles has mean 505 ml. Construct a 95% CI for the mean. *[2 mark(s), Moderate]*

Solution.

$$\text{CI} = 505 \pm 1.96\frac{16}{\sqrt{25}} = 505 \pm 1.96(3.2) = 505 \pm 6.27 = (498.73,\ 511.27).$$

Q5.14. A sample of 10 students has mean 75 and sample standard deviation 5. Construct a 95% CI for the mean. *[2 mark(s), Moderate]*

Solution. Using $t_{0.025,9} = 2.262$:

$$\text{CI} = 75 \pm 2.262\frac{5}{\sqrt{10}} = 75 \pm 3.58 = (71.42,\ 78.58).$$

Q5.15. A sample of 15 items has variance 20. Test whether the population variance is 16 at 5% significance. *[2 mark(s), Moderate]*

Solution.

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{14 \times 20}{16} = 17.5.$$

Compared with $\chi^2_{0.025,14} = 26.12$ and $\chi^2_{0.975,14} = 5.63$, since $5.63 < 17.5 < 26.12$ we fail to reject H_0.

Q5.16. Find the 90th percentile of a χ^2 variable with 9 degrees of freedom. *[2 mark(s), Moderate]*

Solution. $\chi^2_{0.90,9} \approx 14.684$.

Q5.17. Length L is normal with variance 25 mm^2. A customer specifies 100 ± 2.00 mm and rejects parts outside these limits. For $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ with $n = 50$, find the probability of a Type I error. *[8 mark(s), Moderate]*

Solution. $\text{SE} = \dfrac{\sigma}{\sqrt{n}} = \dfrac{5}{\sqrt{50}} = 0.707$. The rejection limits 100 ± 2 correspond to $z = \pm\dfrac{2}{0.707} = \pm 2.83$.

$$\alpha = \text{P}(Z < -2.83) + \text{P}(Z > 2.83) = 0.0023 + 0.0023 = 0.0046 \ (0.46\%).$$

Q5.18. Explain the steps of hypothesis testing with an example. *[8 mark(s), Moderate]*

Solution. **Seven-step procedure** (illustrated for a mean burning rate): (1) identify the parameter of interest (μ); (2) state $H_0 : \mu = 50$; (3) state $H_1 : \mu \neq 50$; (4) choose the test statistic $z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$; (5) set the rejection rule (reject if $|z_0| > z_{0.025} = 1.96$, or if the p-value < 0.05); (6) compute: with $\bar{x} = 51.3, \sigma = 2, n = 25$, $z_0 = 3.25$; (7) conclude: since the p-value $= 2[1 - \Phi(3.25)] = 0.0012 < 0.05$, reject H_0 — the mean burning rate differs from (in fact exceeds) 50 cm/s.

Q5.19. (a) State the hypotheses, statistic, assumptions and rejection region for a two-sided test of a population variance. (b) Find $\text{P}(2.06 < T < 3.45)$ for a t-variable with 25 df. (c) Find x such that a χ^2 variable with 15 df is at most x with 95% probability. *[8 mark(s), Moderate]*

Solution. (a) $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$; statistic $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ (assumes a normal population); reject if $\chi^2 > \chi^2_{\alpha/2,n-1}$ or $\chi^2 < \chi^2_{1-\alpha/2,n-1}$.
(b) $\text{P}(2.06 < T < 3.45) = \text{P}(T < 3.45) - \text{P}(T < 2.06) \approx 0.9990 - 0.9750 = 0.022$.
(c) $\text{P}(\chi^2 \leq x) = 0.95 \Rightarrow x = \chi^2_{0.95,15} \approx 24.996$.

Q5.20. The sensor life is normal. A sample of 30 sensors has mean 800 h and sample standard deviation 70 h. (i) 95% two-sided CI for the mean. (ii) 99% two-sided CI for the variance. (iii) If the population standard deviation is 90 h, a 99% CI for the mean. (iv) Comment on the difference between (ii) and (iii). *[8 mark(s), Moderate]*

Solution. (i) $t_{0.025,29} = 2.045$: $800 \pm 2.045\dfrac{70}{\sqrt{30}} = 800 \pm 26.1 = (773.9,\ 826.1)$.
(ii) $(n-1)s^2 = 29(4900) = 142100$; with $\chi^2_{0.005,29} = 52.34,\ \chi^2_{0.995,29} = 13.12$: $\left(\dfrac{142100}{52.34}, \dfrac{142100}{13.12}\right) = (2715,\ 10831)$.
(iii) $\sigma = 90$ known, $z_{0.005} = 2.576$: $800 \pm 2.576\dfrac{90}{\sqrt{30}} = 800 \pm 42.3 = (757.7,\ 842.3)$.
(iv) The variance CI is far wider because the χ^2 distribution is highly skewed at small samples; the mean CI in (iii) is narrower than (i) because a known σ (using z) gives a more precise estimate than using s (using t).

Q5.21. For the sample $6, 3, 2, 1, 1, 0, 1, 3, 3, 2$ (defects in returned appliances): (i) 90% CI for the mean if $\sigma = 2$. (ii) Sample size to keep the CI width below 2. (iii) 95% CI for the mean if the variance is unknown. *[8 mark(s), Moderate]*

Solution. $\bar{x} = 2.2,\ S_{xx} = 25.6,\ s^2 = 2.844,\ s = 1.687$.
(i) $z_{0.05} = 1.645$: $2.2 \pm 1.645\dfrac{2}{\sqrt{10}} = 2.2 \pm 1.04 = (1.16,\ 3.24)$.
(ii) Width $= 2z\sigma/\sqrt{n} < 2 \Rightarrow \sqrt{n} > 1.645(2) = 3.29 \Rightarrow n > 10.8$, so $n = 11$.
(iii) $t_{0.025,9} = 2.262$: $2.2 \pm 2.262\dfrac{1.687}{\sqrt{10}} = 2.2 \pm 1.21 = (0.99,\ 3.41)$.

Q5.22. Numbers of failed specimens on 10 days are $100, 115, 150, 170, 90, 135, 180, 110, 150, 155$. Assuming normality, test at 10% whether the mean is at most 120. *[8 mark(s), Moderate]*

Solution. $H_0 : \mu \le 120$ vs $H_1 : \mu > 120$, $\alpha = 0.10$, $n = 10$. $\bar{x} = 135.5,\ s = 30.5$.

$$t = \frac{\bar{x} - 120}{s/\sqrt{n}} = \frac{15.5}{30.5/\sqrt{10}} = \frac{15.5}{9.65} = 1.607.$$

Since $t = 1.607 > t_{0.10,9} = 1.383$, we **reject** H_0: there is evidence the mean exceeds 120.

Q5.23. The number of defects in PCBs is hypothesised Poisson. A sample of $n = 60$ boards gives observed frequencies 32, 15, 9, 4 for 0, 1, 2, 3 defects. Perform a goodness-of-fit test. *[8 mark(s), Easy]*

Solution. H_0: defects are Poisson; H_1: not Poisson; $\alpha = 0.05$. Estimated mean $\hat{\lambda} = \dfrac{0(32) + 1(15) + 2(9) + 3(4)}{60} = \dfrac{45}{60} = 0.75$. Expected frequencies $E_i = 60p_i$:

Defects	0	1	≥ 2
Observed	32	15	13
Expected	28.32	21.24	10.44

(The last cells are pooled since the expected count was below 3.)

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{3.68^2}{28.32} + \frac{6.24^2}{21.24} + \frac{2.56^2}{10.44} = 0.478 + 1.833 + 0.628 = 2.94.$$

With df $= 3 - 1 - 1 = 1$, $\chi^2_{0.05,1} = 3.84$. Since $2.94 < 3.84$, we fail to reject H_0; a Poisson model is adequate.

Q5.24. Length L is normal with variance 25 mm^2; test at 10% whether the population variance exceeds 10 using the data $24, 22, 25, 23, 20, 23, 21, 22, 25, 22$. *[8 mark(s), Moderate]*

Solution. $H_0 : \sigma^2 = 10$ vs $H_1 : \sigma^2 > 10$. $\bar{x} = 22.7$, $S_{xx} = 24.1$, $s^2 = \dfrac{24.1}{9} = 2.68$.

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9(2.68)}{10} = 2.41.$$

Since $\chi^2 = 2.41 < \chi^2_{0.10,9} = 14.68$, we fail to reject H_0: there is not enough evidence that the variance exceeds 10. *(The circulated key $s^2 = 3.88, \chi^2 = 3.49$ is incorrect; the correct $s^2 = 2.68$ leaves the conclusion unchanged.)*

Q5.25. Body temperatures of 25 female subjects are 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, 99.0. Test $H_0 : \mu = 98.6$ vs $H_1 : \mu \neq 98.6$ at $\alpha = 0.05$. *[8 mark(s), Moderate]*

Solution. $\bar{x} = 98.264$, $s = 0.4821$, $n = 25$.

$$t = \frac{\bar{x} - 98.6}{s/\sqrt{n}} = \frac{98.264 - 98.6}{0.4821/5} = \frac{-0.336}{0.0964} = -3.49.$$

Since $|t| = 3.49 > t_{0.025,24} = 2.064$, we **reject** H_0: the mean body temperature differs significantly from 98.6°F. *(The circulated key $t = -1.79$, "fail to reject", is incorrect.)*

Q5.26. For the data 24, 22, 25, 23, 20, 23, 21, 22, 25, 22: (i) estimate the population mean, population variance and the variance of the sample mean. (ii) 90% lower bound for the mean. (iii) 95% upper bound for the variance. *[8 mark(s), Moderate]*

Solution. $\bar{x} = 22.7$, $s^2 = 2.68$.

(i) $\hat{\mu} = 22.7$, $\hat{\sigma}^2 = 2.68$, $\mathrm{Var}(\bar{X}) = \dfrac{s^2}{n} = \dfrac{2.68}{10} = 0.268$.

(ii) $t_{0.10,9} = 1.383$: lower bound $= 22.7 - 1.383\dfrac{1.637}{\sqrt{10}} = 22.7 - 0.72 = 21.98$.

(iii) $\chi^2_{0.95,9} = 3.325$: upper bound $= \dfrac{(n-1)s^2}{\chi^2_{0.95,9}} = \dfrac{9(2.68)}{3.325} = 7.25$.

Q5.27. An engineer studying tensile strength (normal) samples 12 specimens with mean 3250 psi and known $\sigma = 60$ psi. (i) Test $H_0 : \mu = 3500$ at 1%. (ii) The smallest significance level at which H_0 is rejected. (iii) Relate the test to a two-sided CI. (iv) How is the test done if σ is unknown? *[8 mark(s), Moderate]*

Solution. (i) $z = \dfrac{3250 - 3500}{60/\sqrt{12}} = \dfrac{-250}{17.32} = -14.43$. Since $|z| = 14.43 > z_{0.005} = 2.576$, reject H_0.

(ii) The p-value $= 2\,\mathrm{P}(Z < -14.43) \approx 0$; H_0 is rejected at essentially any significance level.

(iii) The 99% CI is $3250 \pm 2.576(17.32) = (3205.4,\ 3294.6)$; since 3500 lies outside, H_0 is rejected — consistent with (i).

(iv) With σ unknown, use the t-test $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $n - 1 = 11$ degrees of freedom.

Q5.28. The sugar content of canned peaches is normal. A sample of $n = 10$ cans gives $s = 4.8$ mg. (i) 90% two-sided CI for σ. (ii) 95% upper and lower bounds for σ; compare with (i). *[8 mark(s), Moderate]*

Solution. $(n-1)s^2 = 9(23.04) = 207.36$. $\chi^2_{0.05,9} = 16.919$, $\chi^2_{0.95,9} = 3.325$.
(i) $\sigma^2 \in \left(\dfrac{207.36}{16.919}, \dfrac{207.36}{3.325}\right) = (12.26,\ 62.36)$, so $\sigma \in (3.50,\ 7.90)$ mg.
(ii) 95% upper bound: $\sigma < \sqrt{207.36/3.325} = 7.90$; 95% lower bound: $\sigma > \sqrt{207.36/16.919} = 3.50$. These one-sided 95% bounds coincide with the limits of the two-sided 90% interval (each tail carries 5%).

Q5.29. Rainfall (acre-ft) from 20 seeded clouds is 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6. (i) 95% one-sided lower bound on the mean. (ii) 90% two-sided CI and its width. (iii) Sample size to halve the width. *[8 mark(s), Hard]*

Solution. $n = 20$, $\bar{x} = 26.04$, $s = 4.78$.
(i) $t_{0.05,19} = 1.729$: lower bound $= 26.04 - 1.729\dfrac{4.78}{\sqrt{20}} = 26.04 - 1.85 = 24.19$.
(ii) 90% CI $= 26.04 \pm 1.85 = (24.19,\ 27.89)$; width $= 3.70$.
(iii) Width $\propto 1/\sqrt{n}$, so halving the width needs about 4 times the sample: $n \approx \left(\dfrac{1.729(4.78)}{0.925}\right)^2 \approx 80$.

Q5.30. Amino-acid concentrations (mg/kg) in 8 dog-feed packs are $22, 24, 20, 26, 27, 24, 26, 23$. (i) Test at 5% whether the population variance exceeds 5.5; give the p-value. (ii) Construct 90% upper and lower confidence bounds for the variance. *[8 mark(s), Moderate]*

Solution. $n = 8$, $\bar{x} = 24$, $S_{xx} = 38$, $s^2 = \dfrac{38}{7} = 5.43$.
(i) $H_0 : \sigma^2 = 5.5$ vs $H_1 : \sigma^2 > 5.5$: $\chi^2 = \dfrac{(n-1)s^2}{5.5} = \dfrac{38}{5.5} = 6.91$. Since $6.91 < \chi^2_{0.05,7} = 14.07$, fail to reject H_0; p-value $= \mathrm{P}(\chi^2_7 > 6.91) \approx 0.44$.
(ii) 90% bounds: upper $= \dfrac{38}{\chi^2_{0.95,7}} = \dfrac{38}{2.167} = 17.54$; lower $= \dfrac{38}{\chi^2_{0.05,7}} = \dfrac{38}{14.067} = 2.70$. So $2.70 < \sigma^2 < 17.54$.

5.2 Interval Estimation

Q5.31. Define and interpret a confidence interval with at least two examples. *[8 mark(s), Easy]*

Solution. A confidence interval is a range of values, computed from sample data, that is likely to contain the true population parameter. A 95% CI means that if many samples were taken, about 95% of the resulting intervals would contain the true parameter.
Examples: (1) a 95% CI of $(50.04, 53.96)$ for a mean fill weight; (2) a 90% CI of $(1.16, 3.24)$ for the mean number of defects per appliance.